

Program	Bachelor of Technology (BTech)	Semester - 3
Type of Course	-	
Prerequisite		
Rationale	-	
Effective From A.Y.	2025-26	

Teaching Scheme (Contact Hours)							Examination Scheme						
Lecture	Tutorial	Lab	Theory Credit	Practical Credit	Tutorial Credit	Credit	Theory Marks		Practical Marks		Tutorial Marks		Total Marks
							T-ESE	T-CE	P-ESE	P-CE	T-ESE	T-CE	
3	-	2	3	1	-	4	60	40	60	40	-	-	200

SEE - Semester End Examination, T-CE - Internal Theory, P-CE - Internal Practical

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Introduction to Data Structures Basic Terminology, Classification of Data Structures: Primitive and Non-Primitive, Linear and Non-linear, Operations on Data Structures.	4	10
2	Array Array Representation, Array as an Abstract Data Type, Programming Array in C, Sparse Matrices, Sparse Representations, and its Advantages, Row-measure Order and Column-measure Order representation.	4	10
3	Searching and Sorting Linear Search, Binary Search, Bubble Sort, Insertion Sort, Selection Sort, Radix sort.	4	10
4	Stack and Queue Stack Definition and concepts, Operations on stack, Programming , Stack using Array in C, Prefix and Postfix Notations and their Compilation, Recursion, Tower of Hanoi, Representation of Queue, Operation on Queue, Programming Queue using Array in C. Types of Queue, Applications of Stack & Queue.	7	15
5	Linked List-Part I Dynamic Memory Allocation, Structure in C, Singly Linked List, Doubly Linked List, circular linked list.	4	10
6	Linked List-II and Applications of Linked List Linked implementation of Stack, Linked implementation of Queue, Applications of Linked List.	3	8
7	Trees and Graphs Tree Definition, concepts, and Representation, Binary Tree, Binary Tree Traversals, conversion from general to Binary Tree. Threaded Binary Tree, Heap, Binary Search Tree. Tree for Huffman coding, AVL Trees. 2-3 Trees Graph Definition, Concepts, and Representation, Types of Graphs, Breadth First Search, Depth First Search, Spanning Tree, Kruskal's and Prim's Minimum Cost Spanning Tree Algorithms, Dijkstra's Shortest Path Algorithm.	15	28
8	Hashing The Symbol Table Abstract Data Types, Hash Tables, Hashing Functions, Hash collision Resolution Technique, Linear Probing.	4	10
Total		45	101



Suggested Distribution Of Theory Marks Using Bloom's Taxonomy

Level	Remembrance	Understanding	Application	Analyze	Evaluate	Create
Weightage	7	8	8	7	7	8

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:

C01	Explain and differentiate various data structures, their classifications, and operations for organizing and processing data efficiently.
C02	Implement linear data structures such as arrays, stacks, and queues to solve real-world computational problems
C03	Apply and analyze dynamic data structures like linked lists, trees, and graphs for efficient data manipulation and traversal.
C04	Evaluate and compare the performance of various searching, sorting, and hashing techniques in terms of time and space complexity.
C05	Design appropriate data structure-based solutions for specific problems using suitable algorithms and justify the chosen approach.

CO PO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5
PO - 1					
PO - 2					
PO - 3					
PO - 4					
PO - 5					
PO - 6					
PO - 7					
PO - 8					
PO - 9					
PO - 10					
PO - 11					
PO - 12					

Reference Books

1.	Data Structures using C & C++ By Aaron M. Tanenbaum PHI Learning
2.	An Introduction to Data Structures with Applications (TextBook) By Jean-Paul Tremblay, Paul G. Sorenson Tata McGraw Hill



List of Practical

1. Array	Working with the array. • Write a program to read numbers and store it in array and display it. • Write a program to demonstrate the concept of one-dimensional array • Write a program to insert an element in array. • Write a program to delete an element from an array. • Write a program to add two matrix A and B.. • Write a program to concatenate two strings.
2. Linear Search	Write a program to perform Linear Search.
3. Binary Search	Write a program to perform Binary Search.
4. Bubble sort	Write a program to perform Bubble sort.
5. Selection sort	Write a program to perform Selection sort.
6. Insertion sort	Write a program to perform Insertion sort.
7. Stack using Array	Write a program to implement a stack and perform push, pop operation.
8. linear queue	Write a program to perform the following operations in a linear queue –Addition, Deletion, and Traversing.
9. singly linked list	Write a program to perform the following operations in singly linked list – Creation, Insertion, and Deletion.
10. binary tree	Write a program to create a binary tree and perform – Insertion, Deletion, and Traversal.
11. BFS and DFS	Write a program for traversal of graph (B.F.S., D.F.S.).